

AdaPS-COME: Adaptive Prediction-Set Reliability for Safe Test-Time Adaptation of Vision Transformers

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Abstract. Vision transformers and large visual recognition models are increasingly deployed under corruption shifts, where safe inference requires not only accurate predictions but also reliable behavior when the model is wrong. Test-time adaptation (TTA) can update a model online from unlabeled target data, but this also creates an inference-time safety risk: entropy minimization may sharpen wrong-but-confident predictions and allow unreliable target samples to dominate the update. Conservative objectives such as COME mitigate over-confident collapse by explicitly modeling uncertainty at the loss level, yet they do not fully decide which target samples should be trusted during adaptation. In this paper, we propose AdaPS-COME, a reliability-first safe TTA framework for vision transformers under ImageNet-C corruption shifts. AdaPS-COME estimates sample ambiguity through adaptive prediction-set size and assigns lower adaptation weights to samples whose probability mass is spread over many classes. Combined with a COME-style conservative objective, this design provides sample-level trust control during deployment-time adaptation. Experiments on ImageNet-C with a ViT backbone show that AdaPS-COME maintains competitive accuracy while producing more favorable reliability diagnostics for wrong predictions. In particular, it lowers confidence on wrong predictions, increases uncertainty on wrong predictions, yields stronger uncertainty-based error separation, and becomes more conservative as corruption severity increases.

Keywords: Safe Test-Time Adaptation · Reliable Inference · Vision Transformer · Prediction Set · Uncertainty Estimation · ImageNet-C

1 Introduction

Vision transformers [1] and other large visual models are increasingly used in deployment settings where the test distribution can differ sharply from the clean source distribution. For safe inference, the model should not only predict accurately on easy inputs, but also behave cautiously when corruption, weather, blur, noise, or other shifts make its predictions unreliable. ImageNet-C provides a controlled way to study this problem by applying common corruptions at different severity levels [3]. In fully test-time adaptation, target labels are unavailable and source data are typically inaccessible, so the model must decide how to update itself using only the unlabeled target stream.

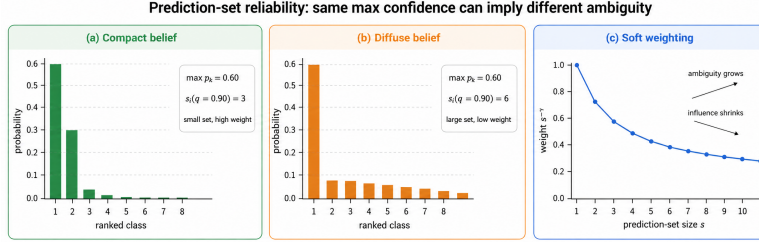


Fig. 1. Prediction-set reliability distinguishes compact and diffuse beliefs with identical maximum confidence, then converts larger prediction sets into smaller reliability weights. The prediction set is a model-induced ambiguity proxy rather than a conformal prediction guarantee.

Test-time adaptation (TTA) can improve robustness by updating the model during deployment, but it also creates a safety risk. Many TTA methods rely on the model’s own predictions as adaptation signals. Tent [8], for example, minimizes prediction entropy on target batches and thereby encourages sharper predictions. EATA [6] and SAR [7] improve stability by filtering unreliable, redundant, or unstable samples. However, under severe corruption, high confidence is not equivalent to correctness. A vision transformer may assign high probability to a wrong class, and adaptation from such wrong-but-confident samples can reinforce confirmation bias. Therefore, confidence is necessary but not sufficient for safe test-time adaptation.

COME [10] addresses this issue from the objective side by introducing conservative uncertainty modeling. Instead of minimizing ordinary softmax entropy alone, COME represents predictions using subjective opinions with explicit uncertainty mass. This makes adaptation more conservative than standard entropy minimization. Nevertheless, COME mainly controls how the loss is formulated; it does not fully solve which target samples should dominate adaptation. In corrupted target batches, uncertain, ambiguous, or confidently wrong samples may still contribute strongly to the update. We argue that conservative TTA requires both an uncertainty-aware objective and a reliability-aware mechanism for controlling sample influence.

The key insight of this work is that reliable target samples should have compact prediction sets. A sample is more trustworthy when most probability mass is concentrated in a small number of classes and its confidence is consistent with low uncertainty. Conversely, if the prediction set is large, the model is spreading probability mass across many candidate classes, which indicates target-sample ambiguity. Such samples should not be treated as equally useful pseudo-supervision signals. This distinction is especially important on ImageNet-C, where increasing corruption severity makes target samples progressively more ambiguous.

We propose AdaPS-COME, an adaptive prediction-set reliability framework for safe test-time adaptation. AdaPS-COME separates the adaptation problem into two coupled questions: what to adapt from and how to adapt conservatively. Prediction-set reliability answers the first question by estimating which target samples are reliable enough to guide adaptation and how strongly they should influence the update. COME-style conservative entropy minimization answers the second question by preventing the model from aggressively collapsing uncertainty. This combination shifts conservative TTA from loss-level uncertainty modeling alone to sample-level reliability-aware adaptation for safer inference under corruption shifts.

Figure 1 makes the motivation concrete. Maximum confidence alone cannot distinguish two distributions that share the same top-class probability but allocate the remaining mass differently. The cumulative-mass prediction set exposes this difference: a compact belief reaches the target mass using only a few classes, whereas a diffuse belief needs many classes and is therefore assigned a smaller adaptation weight.

Our contributions are summarized as follows:

- We propose AdaPS-COME, a reliability-first safe TTA framework that extends conservative TTA from loss-level uncertainty modeling to sample-level trust control.
- We introduce a batch-adaptive prediction-set threshold and a soft reliability weighting mechanism to estimate ambiguity from unlabeled target data beyond maximum confidence.
- We demonstrate on ImageNet-C with ViT that AdaPS-COME maintains competitive accuracy while producing more favorable wrong-prediction reliability and severity-aware conservative behavior.

2 Related Work

2.1 Entropy-Based Test-Time Adaptation

Test-time adaptation updates a pretrained model using unlabeled target data during deployment. Tent [8] is a representative fully test-time adaptation method. It minimizes entropy on target batches and updates a small set of adaptation parameters, such as normalization affine parameters. This design is efficient and source-free, but it assumes that sharpening target predictions is generally beneficial. When the model is confidently wrong, entropy minimization can reinforce incorrect predictions.

Later methods improve TTA stability by controlling which samples or updates are used. EATA [6] performs sample-efficient adaptation by selecting reliable and non-redundant samples while reducing forgetting. SAR [7] studies stable adaptation under dynamic wild-world conditions and avoids unstable updates through sharpness-aware reliability filtering. CoTTA [9] addresses continual test-time adaptation with teacher-student updates and stochastic restoration. DeYO [5] argues that entropy alone is insufficient and introduces transformation-based criteria for selecting or weighting reliable target samples. These methods show that sample reliability is important, but most criteria remain closely tied to entropy, confidence, transformation consistency, or gradient stability rather than explicitly modeling prediction-set ambiguity.

2.2 Safe and Conservative Test-Time Inference

In fully test-time adaptation, the adaptation signal is produced by the deployed model itself. A method that adapts aggressively can appear useful on average while still increasing the influence of wrong-but-confident samples. Safe deployment therefore requires the adaptation mechanism to expose target ambiguity and reduce the effect of unreliable samples on online updates.

COME revisits entropy minimization through a conservative uncertainty-aware formulation [10]. It converts model outputs into subjective opinions with class belief masses and an explicit uncertainty mass. By minimizing opinion entropy rather than softmax entropy alone, COME discourages over-confident collapse. However, COME primarily modifies the adaptation objective and does not fully determine which samples should be trusted. AdaPS-COME complements COME by controlling sample influence before applying the conservative objective.

2.3 Prediction-Set Reliability

Reliability estimation commonly uses maximum confidence, entropy, prediction margin, uncertainty, gradient norm, augmentation consistency, or transformation sensitivity. Hard selection discards samples deemed unreliable, whereas soft weighting keeps all samples but scales their influence.

For each sample, AdaPS-COME measures how many top-ranked classes are needed to cover a target probability mass. Compared with maximum confidence, the resulting prediction-set size reflects the shape of the ranked probability distribution. A compact set indicates concentrated probability mass; a large set indicates ambiguity across many classes. We use prediction set in an operational sense as a top cumulative-mass set, not as a statistically calibrated conformal set. No held-out calibration set, conformal quantile, or finite-sample coverage guarantee is used or claimed.

3 Method

AdaPS-COME separates safe test-time adaptation into two coupled questions. First, *what should the model adapt from?* Adaptive prediction-set reliability estimates target-sample ambiguity without labels. Second, *how should the model adapt once reliability is known?* A weighted conservative objective combines Tent and COME. In short, COME determines how to adapt conservatively, while AdaPS determines which samples should be trusted and how strongly they should influence adaptation.

3.1 Problem Setup

Let f_θ be a source model trained on a clean source distribution $\mathcal{D}_s$. During deployment, the model receives an unlabeled target stream from a shifted distribution $\mathcal{D}_t$. At time step t , a target batch is denoted by $\mathcal{B}_t = \{x_i\}_{i=1}^B$. Target labels are not used for adaptation and are only used for evaluation.

For each sample x_i , the model outputs logits

$$z_i = f_\theta(x_i), \quad (1)$$

and softmax probabilities

$$p_i = \text{softmax}(z_i). \quad (2)$$

We follow the fully test-time adaptation setting and update only a lightweight set of adaptation parameters while keeping the rest of the source model fixed. In the ImageNet-C ViT experiments, the trainable parameters are the affine weights and biases of LayerNorm modules. The classifier head and the remaining transformer parameters are frozen.

3.2 COME Uncertainty Modeling

COME models a prediction as a subjective opinion with class belief masses and an explicit uncertainty mass. The implementation first applies detached logit-norm normalization,

$$\tilde{z}_i = \frac{z_i}{\|z_i\|_2 + \epsilon} \text{sg}(\|z_i\|_2) \tau, \quad (3)$$

where $\text{sg}(\cdot)$ denotes stop-gradient and $\tau = 1$ by default. Given normalized logits $\tilde{z}_{ik}$ for K classes, the default evidence mapping uses non-negative exponential evidence:

$$\alpha_{ik} = \exp(\text{ReLU}(\tilde{z}_{ik})), \quad e_{ik} = \alpha_{ik} - 1. \quad (4)$$

The normalizing strength is

$$D_i = \sum_{k=1}^K \alpha_{ik} = \sum_{k=1}^K e_{ik} + K. \quad (5)$$

The belief mass for class k and the uncertainty mass are

$$b_{ik} = \frac{e_{ik}}{D_i}, \quad u_i = \frac{K}{D_i}. \quad (6)$$

The opinion vector is

$$o_i = [b_{i1}, b_{i2}, \dots, b_{iK}, u_i]. \quad (7)$$

COME minimizes an opinion entropy $H_{\text{COME}}(o_i)$, which preserves explicit uncertainty and is therefore more conservative than minimizing softmax entropy alone. The evidence mapping above is an implementation choice for conservative uncertainty modeling and is not used to claim calibrated uncertainty or coverage guarantees.

3.3 Adaptive Prediction-Set Reliability

A fixed prediction-set threshold, such as $q = 0.95$, applies the same reliability criterion to all target batches. This is undesirable under corruption shifts because target difficulty changes with severity. AdaPS-COME instead computes a batch-adaptive threshold q_t from unlabeled target uncertainty.

First, we estimate batch-level uncertainty by averaging COME uncertainty over the batch:

$$U_t = \frac{1}{B} \sum_{i=1}^B u_i. \quad (8)$$

Since u_i is already bounded by the subjective-logic normalization, the canonical implementation maps this batch uncertainty directly into a bounded threshold:

$$q_t = \text{clip}(q_{\min} + (q_{\max} - q_{\min})U_t, q_{\min}, q_{\max}). \quad (9)$$

In our default setting, $q_{\min} = 0.90$ and $q_{\max} = 0.98$. This direct mapping avoids an additional running min-max normalization state, reducing cold-start and stream-order sensitivity. When the batch appears reliable, q_t is smaller and the prediction set can remain compact. When the batch is uncertain, q_t increases, which leads to larger prediction sets and stronger down-weighting of ambiguous samples.

For each sample, we sort the predicted probabilities in descending order:

$$p_{i(1)} \geq p_{i(2)} \geq \dots \geq p_{i(K)}. \quad (10)$$

The prediction-set size under the adaptive threshold q_t is

$$s_i(q_t) = \min \left\{ s : \sum_{j=1}^s p_{i(j)} \geq q_t \right\}. \quad (11)$$

A small $s_i(q_t)$ means that the model concentrates most probability mass in a few classes, indicating a more reliable target sample. A large $s_i(q_t)$ means that the probability mass is spread across many classes, indicating ambiguity.

We emphasize that this is a model-induced ambiguity proxy, not a conformal prediction procedure. AdaPS-COME does not use a held-out calibration set, does not estimate conformal quantiles, and does not claim finite-sample coverage guarantees. The term prediction set refers only to the top-ranked label set needed to reach the adaptive probability-mass threshold under the model’s current prediction.

3.4 Reliability-Based Adaptation Weight

AdaPS-COME uses prediction-set size to control how strongly each target sample contributes to adaptation. We define a monotone decreasing reliability weight

$$w_i = \phi(s_i(q_t)), \quad \phi'(s) \leq 0. \quad (12)$$

The default implementation uses inverse-square-root weighting,

$$w_i = (s_i(q_t))^{-\gamma}, \quad (13)$$

where γ controls the strength of reliability weighting and is set to 0.5 by default. The weights are detached from the computation graph before multiplying the adaptation loss, preventing degenerate gradients through the reliability estimator.

This weighting is deliberately soft rather than purely selective. High-reliability samples receive larger weights, while ambiguous samples remain in the batch but have reduced influence. This avoids the brittleness of hard thresholding and allows adaptation to use all available target data with reliability-aware gradient scaling.

3.5 Weighted Conservative Adaptation Objective

The standard Tent entropy loss for a sample is

$$H(p_i) = - \sum_{k=1}^K p_{ik} \log p_{ik}. \quad (14)$$

AdaPS-COME combines Tent entropy with COME opinion entropy:

$$\ell_i = (1 - \lambda)H(p_i) + \lambda H_{\text{COME}}(o_i), \quad (15)$$

where λ controls the contribution of conservative uncertainty modeling. The final weighted objective is

$$\mathcal{L}_{\text{AdaPS}} = \frac{\sum_{i=1}^B w_i \ell_i}{\sum_{i=1}^B w_i + \epsilon}. \quad (16)$$

The model is updated by

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}_{\text{AdaPS}}. \quad (17)$$

Only the selected adaptation parameters are updated. The classifier and frozen source representation remain unchanged unless otherwise specified by the backbone-specific protocol.

4 Experiments

We evaluate AdaPS-COME under ImageNet-C corruption shifts with a ViT backbone. The evaluation asks four questions: (i) whether AdaPS-COME maintains competitive accuracy compared with the direct COME-style baseline, (ii) whether it reduces over-confidence on wrong predictions, (iii) whether uncertainty better separates correct and incorrect predictions, and (iv) whether prediction-set reliability becomes more conservative as corruption severity increases.

4.1 Dataset, Baselines, and Protocol

The main evaluation uses ImageNet-C with corruption severities 1–5 and a ViT-Base backbone. The source model is trained on clean ImageNet, and target images are presented without labels during adaptation. Labels are used only for evaluation after each target batch. Tent represents aggressive entropy minimization; EATA represents entropy-based adaptation with reliability and redundancy filtering; SAR represents stable sharpness-aware adaptation. Tent-COME is the direct COME-style baseline. AdaPS-COME adds sample-level prediction-set reliability on top of the conservative COME objective.

Adaptation updates only LayerNorm affine parameters, keeps dropout disabled by running the model in evaluation mode, and uses seed 0, batch size 64, one online update per target batch, learning rate 10^{-3} , SGD with momentum 0.9, no weight decay, $\lambda = 0.03$, COME p -norm 2, $q_{\min} = 0.90$, $q_{\max} = 0.98$, and inverse-square-root prediction-set weighting. The target stream is processed in the recorded corruption and severity order without shuffling. Metrics are computed with the current model before applying the unlabeled update on the same batch.

4.2 Accuracy and Safe-Inference Metrics

Table 1 reports severity-wise top-1 accuracy, and Figure 2 visualizes the same trends across the Tent, EATA, and SAR families. The highest absolute accuracy is obtained by SAR+COME. AdaPS-COME is not designed to maximize top-1 accuracy by forcing all target predictions to become sharper; it is evaluated as a reliability-first method that controls unreliable sample influence.

Table 1. Average top-1 accuracy on ImageNet-C using ViT-Base across corruption severities 1–5. Values are averaged over 19 corruption types. Bold indicates the highest value at each severity.

Method	Severity 1	Severity 2	Severity 3	Severity 4	Severity 5
Tent	72.12	66.20	62.05	54.95	44.41
Tent+COME	72.20	66.29	62.14	55.04	44.50
Tent+AdaPS-COME	71.67	65.58	61.38	54.31	43.95
EATA	72.39	67.51	64.30	58.88	51.61
EATA+COME	72.34	67.68	64.41	59.44	52.17
EATA+AdaPS-COME	71.84	66.72	63.37	58.34	50.00
SAR	75.55	71.86	69.39	64.66	56.30
SAR+COME	75.67	72.07	69.55	64.75	56.72
SAR+AdaPS-COME	74.53	70.88	68.32	63.56	55.89

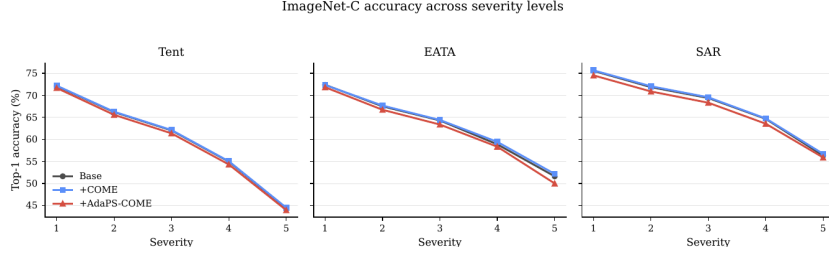


Fig. 2. Top-1 accuracy across ImageNet-C severity levels for the Tent, EATA, and SAR families. Each panel compares the base method, its COME variant, and its AdaPS-COME variant. The nearly parallel trends make the reliability–accuracy trade-off explicit rather than suggesting an accuracy-dominance claim.

Top-1 accuracy measures classification performance and expected calibration error (ECE) measures confidence calibration. Because safe inference depends on how the model behaves when it is wrong, we additionally report wrong-prediction confidence and wrong-prediction uncertainty:

$$\text{WrongConf} = \mathbb{E} \left[\max_k p_{ik} \mid \hat{y}_i \neq y_i \right], \quad (18)$$

$$\text{WrongUnc} = \mathbb{E}[u_i \mid \hat{y}_i \neq y_i]. \quad (19)$$

Lower wrong confidence and higher wrong uncertainty are desirable because wrong predictions should not be treated as reliable pseudo-supervision. We also report AUROC_u , the AUROC obtained by using uncertainty to distinguish wrong predictions from correct predictions. Confidence separation and uncertainty separation are defined as

$$\text{ConfSep} = \mathbb{E} \left[\max_k p_{ik} \mid \hat{y}_i = y_i \right] - \mathbb{E} \left[\max_k p_{ik} \mid \hat{y}_i \neq y_i \right], \quad (20)$$

$$\text{UncSep} = \mathbb{E}[u_i \mid \hat{y}_i \neq y_i] - \mathbb{E}[u_i \mid \hat{y}_i = y_i]. \quad (21)$$

Larger separation indicates that confidence and uncertainty are more aligned with prediction correctness.

4.3 Main Results: Competitive Accuracy and Wrong-Prediction Reliability

Table 2 directly compares Tent-COME and AdaPS-COME. AdaPS-COME has a modest accuracy decrease of 0.65 percentage points, while ECE changes by only 0.04 points. We therefore do not claim an average accuracy or ECE improvement. The main advantage is visible in wrong-prediction reliability.

In Table 3, we evaluate reliability using five metrics. **WC** denotes wrong-prediction confidence, where a lower value means that incorrect predictions are less over-confident. **WU** denotes wrong-prediction uncertainty, where a higher value means that the model better recognizes its errors. **AUROC** measures how well uncertainty separates incorrect predictions from correct ones. **CS** and **US** denote confidence separation and uncertainty separation, respectively; higher values indicate better distinction between correct and incorrect predictions.

AdaPS-COME reduces wrong-prediction confidence from 0.451 to 0.446. This means that when the model is wrong, it is less confident in the incorrect prediction. It also

Table 2. Reliability-focused ImageNet-C results with a ViT backbone. Accuracy and ECE are reported in percentage. Higher AUROC_u, confidence separation, and uncertainty separation are better. Lower wrong confidence and higher wrong uncertainty are better.

Method	Acc.	ECE	Wrong Conf.	Wrong Unc.	AUROC _u	Conf. Sep.	Unc. Sep.
Tent-COME	60.03	4.98	0.451	0.117	0.712	0.312	0.043
AdaPS-COME	59.38	5.02	0.446	0.119	0.715	0.315	0.045

Table 3. Reliability ablation across three TTA frameworks on ImageNet-C with ViT-Base. Metrics are averaged over all severities and corruptions. Deltas are computed relative to each family-specific base TTA method. Bold indicates the most favorable reliability value within each family.

Family	Variant	WC↓	ΔWC↓	WU↑	ΔWU↑	AUROC↑	ΔAUC↑	CS↑	ΔCS↑	US↑	ΔUS↑
Tent	Base	0.451	–	0.116	–	0.712	–	0.312	–	0.043	–
	+COME	0.451	-0.000	0.117	+0.001	0.712	+0.001	0.312	-0.000	0.043	+0.000
	+AdaPS-COME	0.446	-0.005	0.119	+0.002	0.715	+0.003	0.315	+0.003	0.045	+0.002
EATA	Base	0.524	–	0.061	–	0.706	–	0.312	–	0.033	–
	+COME	0.523	-0.001	0.062	+0.001	0.706	+0.000	0.312	-0.000	0.033	+0.000
	+AdaPS-COME	0.512	-0.011	0.067	+0.006	0.714	+0.008	0.317	+0.005	0.037	+0.003
SAR	Base	0.591	–	0.033	–	0.723	–	0.286	–	0.019	–
	+COME	0.591	-0.000	0.033	+0.000	0.721	-0.001	0.286	+0.000	0.019	+0.000
	+AdaPS-COME	0.572	-0.019	0.039	+0.006	0.735	+0.012	0.298	+0.012	0.023	+0.005

increases wrong-prediction uncertainty from 0.117 to 0.119, indicating that the model better recognizes unreliable predictions rather than forcing every target sample to become sharp. The uncertainty-based error separation moves in the desired direction: AUROC_u increases from 0.712 to 0.715, confidence separation from 0.312 to 0.315, and uncertainty separation from 0.043 to 0.045.

The cross-family ablation in Table 3 strengthens the reliability-first interpretation. Across Tent, EATA, and SAR, AdaPS-COME consistently moves reliability metrics in the desirable direction, although the numerical margins are modest. Thus, the main empirical advantage is not a higher average accuracy value, but a more useful confidence–uncertainty structure for identifying unreliable target samples.

4.4 Severity-Aware Conservative Behavior

Table 4 gives the detailed Tent-family severity diagnostics. As corruption severity increases from 1 to 5, accuracy decreases from 71.67 to 43.95 and ECE increases from 2.33 to 8.60, confirming that the target distribution becomes progressively harder.

Table 4. Severity diagnostics for AdaPS-COME on ImageNet-C with a ViT backbone. Set Size denotes average prediction-set size, and Weight denotes average normalized reliability weight used for adaptation diagnostics.

Severity	Acc.	ECE	Wrong Conf.	Wrong Unc.	Set Size	Weight
1	71.67	2.33	0.505	0.093	46.9	0.541
2	65.58	3.68	0.473	0.106	66.3	0.465
3	61.38	4.53	0.452	0.116	82.4	0.417
4	54.31	5.99	0.419	0.130	108.3	0.346
5	43.95	8.60	0.381	0.149	148.9	0.263

The prediction-set mechanism tracks this increasing difficulty. Average prediction-set size grows from 46.9 to 148.9, while the average adaptation weight decreases from

0.541 to 0.263. Thus, AdaPS-COME automatically reduces the influence of target samples when they become less reliable. Wrong-prediction behavior follows the same conservative pattern: wrong confidence decreases from 0.505 to 0.381, while wrong uncertainty increases from 0.093 to 0.149. Figure 3 and Table 5 show that the same monotonic response holds across Tent, EATA, and SAR, rather than being specific to one base adaptation method.

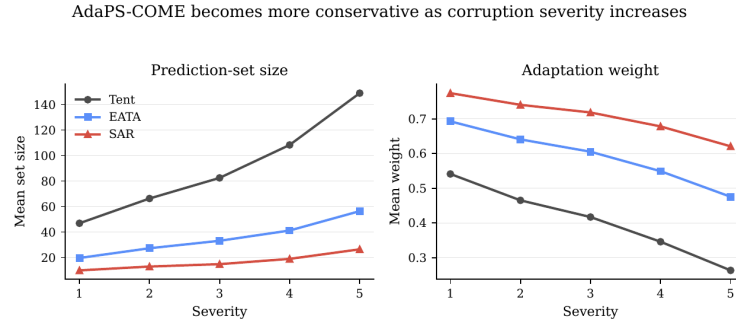


Fig. 3. Cross-family severity diagnostics. Higher corruption severity produces larger prediction sets and lower adaptation weights for Tent+AdaPS-COME, EATA+AdaPS-COME, and SAR+AdaPS-COME. The shared monotonic response supports a family-independent conservative-control interpretation.

Table 5. Severity-aware AdaPS-COME diagnostics across the Tent, EATA, and SAR adaptation families. Larger prediction sets and smaller weights indicate more conservative adaptation under higher target ambiguity; lower wrong confidence and higher wrong uncertainty are desirable.

Family	Metric	Sev. 1	Sev. 2	Sev. 3	Sev. 4	Sev. 5
Tent	Set Size↑	46.9	66.3	82.4	108.3	148.9
Tent	Weight↓	0.541	0.465	0.417	0.346	0.263
Tent	Wrong Conf.↓	0.505	0.473	0.452	0.419	0.381
Tent	Wrong Unc.↑	0.093	0.106	0.116	0.130	0.149
EATA	Set Size↑	19.6	27.3	33.1	41.2	56.3
EATA	Weight↓	0.693	0.641	0.605	0.549	0.476
EATA	Wrong Conf.↓	0.556	0.532	0.516	0.493	0.465
EATA	Wrong Unc.↑	0.052	0.059	0.065	0.073	0.084
SAR	Set Size↑	9.9	12.9	14.8	18.9	26.4
SAR	Weight↓	0.774	0.740	0.718	0.678	0.621
SAR	Wrong Conf.↓	0.604	0.586	0.576	0.558	0.538
SAR	Wrong Unc.↑	0.032	0.036	0.038	0.042	0.048

Table 5 further connects the internal reliability controller to the behavior of incorrect predictions. Across all three TTA families, increasing corruption severity is associated with lower confidence and higher uncertainty on wrong predictions. For example, in the Tent family, wrong-prediction confidence decreases from 0.505 to 0.381, while wrong-prediction uncertainty increases from 0.093 to 0.149. EATA and SAR show the same directional pattern, although their absolute values differ.

4.5 Evidence Boundary and Pending Ablations

The current aggregate artifacts support wrong-confidence, wrong-uncertainty, separation metrics, AUROC, runtime, memory, and batch-level prediction-set diagnostics. They do not contain raw per-sample logits, probabilities, or prediction records. Therefore, high-confidence error rate, selective-risk curves, AURC/E-AURC, and thresholded abstention analyses cannot be reconstructed reliably from the current package.

The most important pending studies are comparisons with scalar weighting signals such as confidence, entropy, margin, and uncertainty weighting; fixed-threshold versus adaptive prediction sets; sensitivity to λ , $q_{\min}$, $q_{\max}$, and γ ; multi-seed variance; and per-corruption confidence intervals. These should be reported using the same reliability metrics rather than accuracy alone.

5 Discussion

The results suggest that safe test-time adaptation should not be evaluated only by average accuracy. In unlabeled adaptation, a method can increase accuracy by aggressively sharpening predictions, but this can be unsafe if wrong predictions also become highly confident. AdaPS-COME takes a reliability-first view: it accepts a small accuracy trade-off relative to Tent-COME while improving the uncertainty structure of wrong predictions.

This distinction is important under severe corruption. When images are heavily corrupted, the model’s probability mass naturally becomes less concentrated. AdaPS-COME detects this through larger prediction sets and reduces adaptation weight accordingly. The signal is not reducible to maximum confidence because samples with the same top probability may distribute the remaining mass very differently.

The results also clarify the complementary roles of COME and AdaPS. COME provides a conservative objective, but the objective alone does not decide which samples should guide adaptation. AdaPS-COME adds this missing sample-level reliability control. The contribution is therefore not merely another entropy term or threshold; it is the connection between prediction-set ambiguity and conservative adaptation strength.

5.1 Limitations and Scope

AdaPS-COME is a reliability-control mechanism rather than an accuracy-maximization method. The reliability improvements are consistent across Tent, EATA, and SAR but numerically modest. The main results rely on seed-0 aggregate logs. Prediction sets are model-induced ambiguity proxies, not statistically calibrated conformal prediction sets. Runtime values are implementation-level logs rather than hardware-independent benchmarks. Stronger claims require the pending repeated runs, alternative weighting baselines, sensitivity studies, and raw per-sample selective-risk analyses described above.

6 Conclusion

We presented AdaPS-COME, a reliability-guided safe test-time adaptation method for vision transformers under ImageNet-C corruption shifts. Unlike entropy-minimization methods that update from all target samples with similar influence, AdaPS-COME estimates adaptive prediction-set reliability and reduces the contribution of ambiguous

samples during adaptation. Combined with COME-style uncertainty modeling, this enables conservative adaptation at both the objective level and the sample-weighting level.

Our analysis shows that AdaPS-COME yields a more favorable confidence–uncertainty balance for wrong predictions. Although it slightly decreases average accuracy compared with Tent-COME, it lowers confidence on wrong predictions, increases uncertainty on wrong predictions, and strengthens uncertainty-based separation between correct and incorrect predictions in the recorded seed-0 logs. Severity diagnostics further show that AdaPS-COME enlarges prediction sets and reduces adaptation weights as corruption severity increases. These results suggest that safe TTA should not only optimize how the model adapts, but also explicitly decide which unlabeled target samples are reliable enough to guide adaptation.

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